DO NOW — RECALL

CPU components and registers

Name: _____ Class: _____ Date: _____

1. Which data structure supports last-in-first-out Undo?

2. What is the extra space complexity of allocating a new length-n array?

3. What does the ALU perform?

4. Simplify (A ∧ ¬B) ∨ (A ∧ B) and explain why the grouped cells make one variable disappear.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A stack.

2. Model answer

O(n).

3. Model answer

Arithmetic and logical operations.

4. Model answer

Factor A: A∧(¬B∨B)=A∧1=A. K-map adjacent cells differ only in B, so B is eliminated.

Vocabulary — know these terms

ALU — Arithmetic Logic Unit.

CU — Control Unit, coordinates and decodes instruction operations.

MAR — Memory Address Register.

MDR — Memory Data Register.

CIR — Current Instruction Register.

PC — Program Counter.

ACC — Accumulator, stores intermediate arithmetic results.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Explain the answer

CIR now contains ADD 10; for sequential execution PC has advanced to 201.

IMPROVE: Explain one correction to your own reasoning.

NEW LEARNING

Read, annotate and explain

Understand the CPU areas

The control unit decodes and coordinates operations; the ALU performs arithmetic and logical operations.

Separate address from data

MAR holds the address being accessed; MDR holds the instruction or data transferred; PC holds the next instruction's address.

Trace the simplified fetch

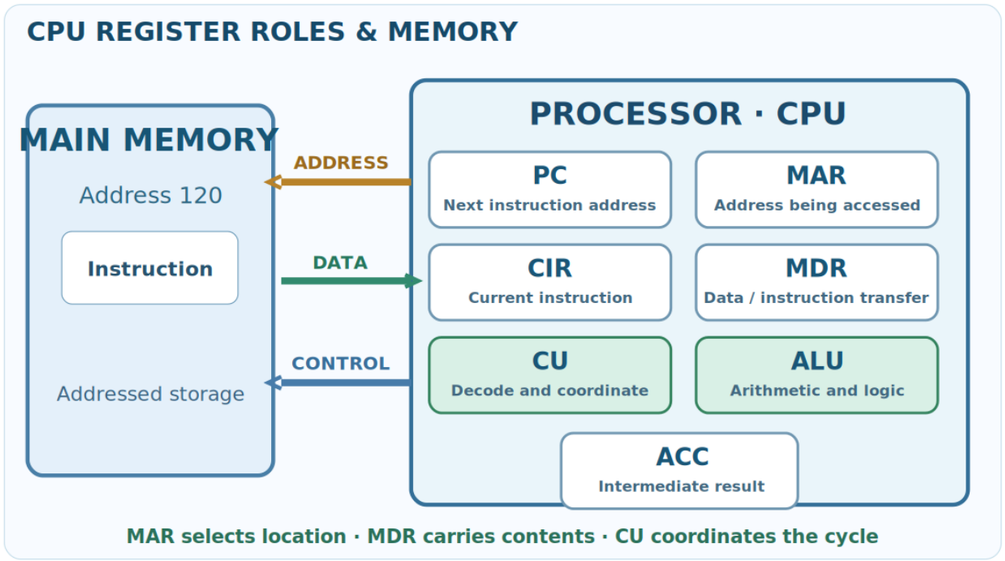
With PC=120, MAR becomes 120; memory supplies instruction via MDR to CIR; PC advances to the next sequential address. Branching may change PC later.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Memory and CPU. Program Counter gives the address of next instruction; MAR holds address; MDR transfers content; CIR holds instruction; control unit coordinates; ALU operates; accumulator stores results. Address, data and control buses connect memory and CPU.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

The MAR stores the contents of the memory location being fetched.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

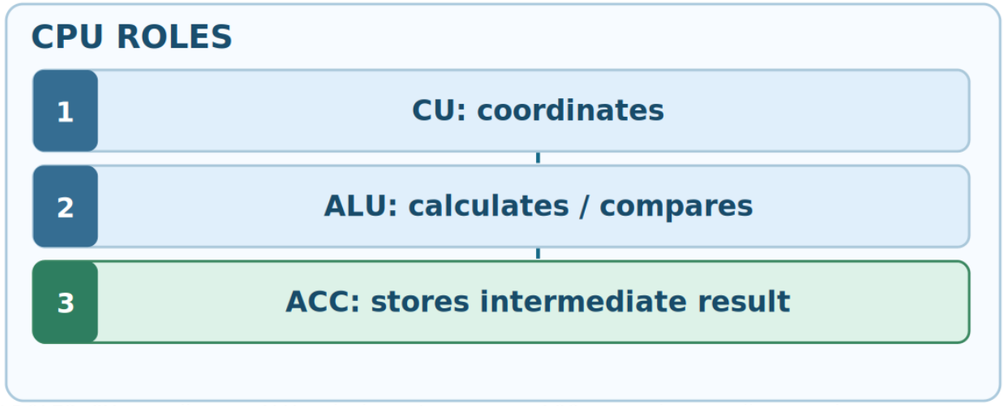
Question 3

Explain why the CPU uses both MAR and MDR rather than a single undifferentiated register, referring to an instruction fetch and a data load.

New Learning · original lesson diagrams

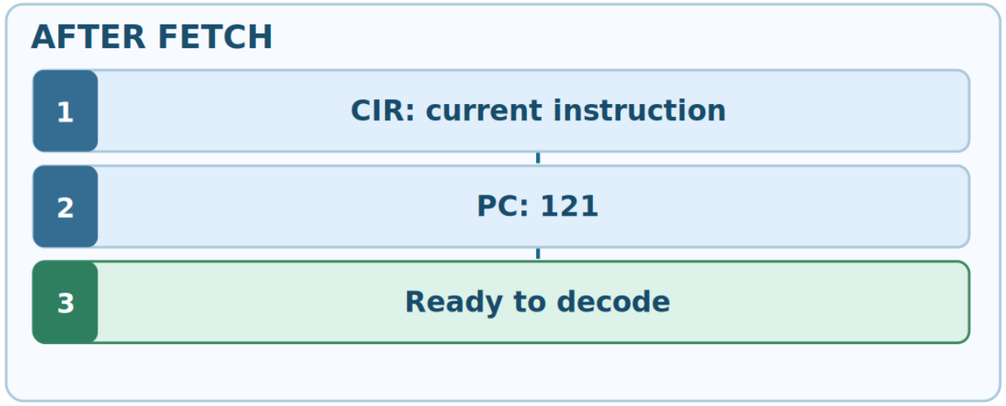
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



CPU ROLES; CU: coordinates; ALU: calculates / compares; ACC: stores intermediate result

New Learning



AFTER FETCH; CIR: current instruction; PC: 121; Ready to decode

I DO — WORKED EXAMPLE

Follow the worked method

If PC starts at address 50, trace register roles for the next sequential instruction fetch.

Predict first

Copy PC=50 into MAR to identify the memory address being accessed.

Explain the answer

Fetch memory content into MDR, copy it to CIR and advance PC to 51.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

State the roles of PC, MAR, MDR, CIR and ACC.

Question 2

Trace a fetch starting PC=73 where Memory[73] contains LDA 92. State each register's role and the sequential next PC.

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first MAR must contain 200 and MDR transports the instruction from memory.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

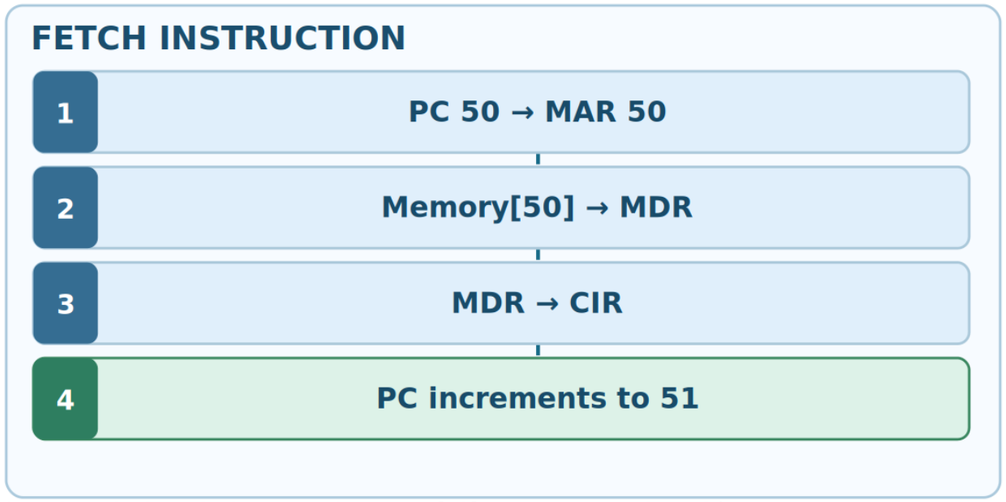
IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

If PC starts at address 50, trace register roles for the next sequential instruction fetch.

Copy PC=50 into MAR to identify the memory address being accessed.



FETCH INSTRUCTION; PC 50 → MAR 50; Memory[50] → MDR; MDR → CIR; PC increments to 51

WE DO — GUIDED PRACTICE

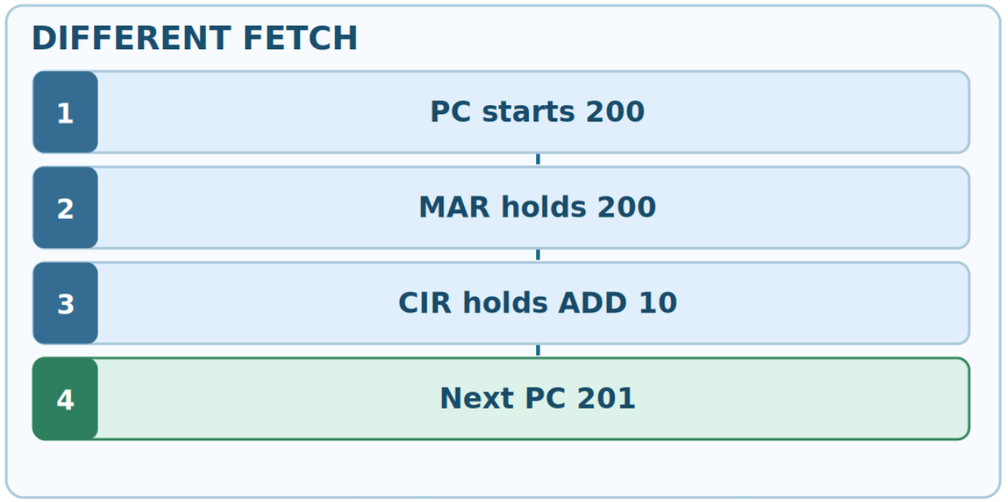
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

If PC initially holds 200 and Memory[200] contains ADD 10, state MAR, CIR and next sequential PC.

MAR must contain 200 and MDR transports the instruction from memory.



DIFFERENT FETCH; PC starts 200; MAR holds 200; CIR holds ADD 10; Next PC 201